Employee Management System (Cloud Based)

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1. Project Title

Cloud Based Employee Management System using Java Full Stack

2. Objective

The objective of this project is to develop a cloud-based employee management system that allows organizations to manage employee records efficiently.
The system provides CRUD operations, search, pagination, validation, and scalable backend architecture.

3. Target Users

• HR Managers
• Admin Staff
• Company Management
• Recruiters

4. System Modules

- **Employee Module**
  - Add employee
  - Update employee
  - Delete employee
  - View employees
  - Search employees
  - Pagination
- **System Module**
  - Global Exception Handling
  - Validation
  - DTO Mapping
  - REST API Integration
- **Future Module**
  - JWT Authentication

- AI Resume Analyzer
- Cloud Deployment
- Swagger API Docs

5. Technology Stack

Backend

- Java
- Spring Boot
- Hibernate JPA

Database

- PostgreSQL (Supabase / Cloud)

Frontend

- React.js
- Axios
- CSS

Cloud

- AWS / Supabase

AI (Planned)

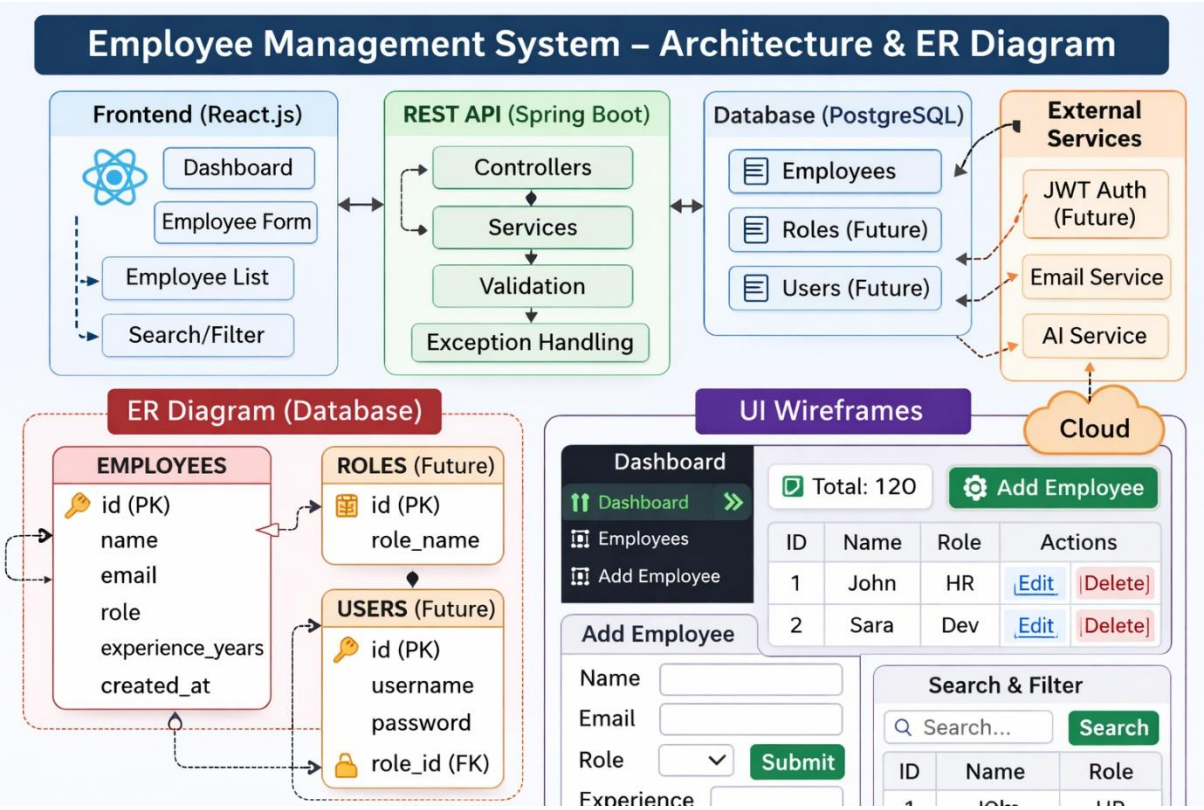
- Resume Skill Matching
- Employee Analytics

6. Functional Workflow

1. User opens application
2. User views employee dashboard
3. User adds new employee
4. Data sent to backend via REST API
5. Backend validates data
6. Data stored in PostgreSQL cloud DB
7. User can update/delete/search employees
8. System handles errors globally

7. Key Features

✓ Cloud Database Integration
✓ RESTful Architecture
✓ Layered Backend Design
✓ DTO Pattern Implementation
✓ Pagination & Search
✓ Exception Handling
✓ Scalable Architecture
✓ Industry-Level Code Structure



8. System Architecture

Frontend (React)
⬇
REST API (Spring Boot)
⬇
Service Layer
⬇
Repository Layer
⬇
PostgreSQL Cloud Database

9. Future Enhancements

- JWT Security
- Role Based Access
- Resume AI Analyzer
- Docker Deployment
- Microservices Migration
- CI/CD Pipeline

10. **Expected Outcome**

The system will provide an efficient cloud-based employee management solution that demonstrates full-stack development capability and industry-ready software architecture skills.